

# HPC School



## HPC Primer - SSH

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# SSH

The Slides in PDF:

<https://edu.nl/98drq>

# **whoami** Instructors

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- Amateur car mechanic.

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- Seasoned systems administrator. Manager of TU/e's Umbrella Cluster.
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# Requirements I

**A SURF Cua account**

<https://portal.cua.surf.nl>

**A terminal on Snellius**

<https://ondemand.snellius.surf.nl>

**Basic terminal (bash) knowledge**

# SSH : Secure Shell

A protocol for securely accessing and managing remote systems over a network.

For using HPC/Supercomputers via the command line (CLI), for example from your laptop

Access to online Cloud services like [github.com](https://github.com).

# Why SSH

- Verify the Identity of remote systems
- Verify the Identity of the user by the remote system
- Encrypt network traffic
- Password less login

# Asymmetric Encryption

## public key

- Used for encryption that only private key can decrypt
- Can be given (published) to everyone.

# private key

- Remains (only) with the user
  - The possession of this key is proof of the user's identity
- Used to generate public key.
- Used to decrypt what was encrypted with the public key
- Password Protected! (not SHOULD, MUST)

# Asymmetric Encryption



# Generating a key pair (CLI)

**ssh-keygen** : generate unique Pub/Priv Key

```
$ ssh-keygen -t ed25519 -C "<EMAIL>"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/Users/alain/.ssh/id_ed25519): <ENTER>
Enter passphrase for "/Users/alain/.ssh/id_ed25519" (empty for no passphrase): <PASSWORD>

$ ls -l $HOME/.ssh/

-rw----- 1 alain  staff  1766 Sep 28  2026 id_ed25519
-rw----- 1 alain  staff   394 Sep 28  2026 id_ed25519.pub
```

# Generating a key pair walk trough

<https://supercomputing.tue.nl/documentation/steps/access/ssh/>

( <https://edu.nl/p4vtc> )

# Using your key (CLI)

`ssh-copy-id` (Linux/Mac): place public key on the correct place on remote system

On remote system key is automatically added to:

```
$HOME/.ssh/authorized_keys
```

## Examples (Linux)

```
ssh-copy-id <username>@snellius.surf.nl  
ssh-copy-id <username>@hpc.tue.nl
```

Then

```
ssh <username>@snellius.surf.nl  
ssh <username>@hpc.tue.nl
```

No longer a password asked (but maybe for your private key)

## Loading your private key into memory

```
ssh-agent bash  
ssh-add
```

The password for your private keys is asked,  
but after that the loaded key is used,  
as long as your session lasts

scp : copy a file to a remote system using SSH

```
scp .ssh/id_ed25519.pub <name>@hpc.tue.nl:~/.ssh/authorized_keys
```

# SSH "one-shot" commands

```
# Check the files in your home directory on snellius
```

```
$ ssh tesr52960@snellius.surf.nl ls -l
```

```
total 70
```

|            |   |           |           |      |     |    |       |                      |
|------------|---|-----------|-----------|------|-----|----|-------|----------------------|
| drwxr-x--- | 2 | tesr52960 | tesr52960 | 4096 | Oct | 8  | 2025  | bin                  |
| drwxr-x--- | 4 | tesr52960 | tesr52960 | 4096 | Oct | 10 | 2024  | snellius-hybrid-jobs |
| drwxr-xr-x | 2 | tesr52960 | tesr52960 | 4096 | Jan | 15 | 2025  | Desktop              |
| drwxr-xr-x | 3 | tesr52960 | tesr52960 | 4096 | Mar | 9  | 2026  | Documents            |
| drwxr-xr-x | 2 | tesr52960 | tesr52960 | 4096 | Jan | 15 | 2025  | Downloads            |
| -rw-r----- | 1 | tesr52960 | tesr52960 | 1550 | Apr | 10 | 16:01 | fluent-950150.trn    |
| -rw-r----- | 1 | tesr52960 | tesr52960 | 6028 | Apr | 13 | 13:50 | fluent103193.trn     |
| drwxr-x--- | 2 | tesr52960 | tesr52960 | 4096 | Nov | 11 | 2025  | GIT                  |

# Creating \$HOME/.ssh/authorized\_keys

```
ssh curxxxx@snellius.surf.nl # Password curxxxx?
```

```
cd .ssh  
cp id_ed25519.pub authorized_keys
```

```
exit  
ssh curxxxx@snellius.surf.nl # Password private key?
```

# We still need a password!

```
exit  
ssh-agent bash  
ssh-add
```

# And now we don't:

```
ssh curxxxx@snellius.surf.nl  
exit
```

```
cat id_ed25519.pub  
cd ..
```

# SSH Assignment I

## Hands on

1. Goto. <https://ondemand.snellius.surf.nl>
2. Open a terminal on Snellius

SURF



Module Browser



&gt;\_ Snellius Shell Access



System Status

OnDemand provides an integrated, single access point

## Pinned Apps A featured subset of [all](#)

### Interactive Apps

3. Create a ssh key public/private keypair
4. Copy the Public key to [test@hpcschool.nl](mailto:test@hpcschool.nl) (T3stT3st%)
5. Check what files are there

# SSH Assignment Solution

```
$ ssh-keygen -t ed25519 -C "Test Key"  
$ ssh-copy-id test@hpcschool.nl  
$ bash ssh-agent  
$ ssh-add  
$ ssh test@hpcschool.nl ls -l
```

# SSH Assignment II

1. Copy Job1.sbatch from hpcschool.nl
2. Copy Job1.sbatch to your local machine from Snellius

# SSH Assignment Solution

```
scp test@hpcschool.nl:~/Job1.sbatch .  
Download via ondemand.snellius.surf.nl (files menu)
```